

MABROUK LOGBIBI

Final-year Undergraduate Student (Computer Science & Automation) Algeria

Email: mabrouk.logbibi@gmail.com

GitHub: <https://github.com/Islamlogbibi>

LinkedIn: <https://www.linkedin.com/in/mabrouk-logbibi-0651b026a/>

Portfolio: <https://new-portfolio-wheat-two.vercel.app/>

Profile / Research Interests

Final-year undergraduate student in a dual Bachelor's program in Computer Science and Automation, ranked 5th nationally in Algeria. Strong interest in Brain-Computer Interfaces (BCI), EEG signal processing, and intelligent human-machine interaction.

Currently focusing on EEG-based motor intention decoding using machine learning, with the goal of developing assistive systems for individuals with motor impairments. Experienced in software development, signal processing, and real-world system deployment.

Education

Bachelor's Degree (Dual Program): Computer Science & Automation

Badji Mokhtar University, Annaba, Algeria

Expected Graduation: June-July 2026

- Ranked 5th nationally within the first cohort of this program
- GPA: 12.56 / 20
- Interdisciplinary training in computer science, automation, and intelligent systems

Relevant Coursework:

Programming (C, C++, Python), Data Structures & Algorithms, Artificial Intelligence, Signal Processing, Control Systems, Automation & Industrial Systems, Operating Systems, Databases

Technical Skills

Programming Languages: Python, C, C++, JavaScript, PHP, SQL

Frameworks & Tools: FastAPI, Django, React, Flutter, OpenCV

AI / Signal Processing: Machine learning (introductory), time-series analysis, EEG signal processing (learning), feature extraction

Systems & Platforms: Linux (Arch, Kali), Git/GitHub, Docker

Databases: MySQL, PostgreSQL, MongoDB

Research & Projects

EEG-Based Motor Intention Decoding (Proposed Research)

- Designing a system to decode motor imagery EEG signals into control commands
- Applying signal preprocessing, feature extraction, and deep learning models (CNN/LSTM)
- Targeting real-time assistive human-computer interaction for individuals with motor impairments

A Multi-Modal Wearable System for Real-Time Sign Language Translation and Adaptive Remote Communication

- Designing a wearable system combining glove-based sensing and EMG signals for robust sign language recognition
- Developing a multi-modal fusion pipeline to improve accuracy under sensor noise and variability
- Integrating real-time translation to text/speech for remote communication scenarios (e.g., calls)
- Exploring adaptive models to personalize recognition based on user-specific patterns

Blood Donation Management System

- Designed and implemented a database-driven system for hospital use
- Deployed on the local server of Annaba Central Hospital and actively used by staff
- Developed backend logic, database structure, and user interface integration

Blood Donation Management System

- Designed and implemented a database-driven system for hospital use
- Deployed on the local server of Annaba Central Hospital and actively used by staff
- Developed backend logic, database structure, and user interface integration

Smart Blood Supply Platform

- Developed a platform connecting healthcare providers with logistics systems for blood supply management
- Achieved startup certification with patent recognition
- Improved coordination of blood bank logistics and transfusion workflows

AI / NLP Experiments

- Conducted experiments with language models and evaluation metrics (ROUGE, perplexity, faithfulness)

- Applied AI models in practical systems and evaluated performance in real-world scenarios
-

Professional Experience

Part-Time Remote Developer — PromptLine (France)

- Contributed to backend development and deployment
 - Worked with production systems, debugging, and collaborative development workflows
-

Achievements

- Ranked 5th nationally in Computer Science & Automation dual program
 - First place in a national competitive programming competition in Algeria
 - 5th place in an international hackathon (Algeria & Tunisia)
 - Patent recognition for Smart Blood Supply platform
 - Certificate from Algerian Olympiad in Informatics (Python Camp)
-

Languages

- Arabic: Native
 - English: Working proficiency (academic and technical use)
 - French: Basic
 - Japanese: Beginner (N5) (actively studying)
-

Leadership & Activities

- President, Research & Development Department — HTIC
 - Former Lead of the CSA Kernel Team within the Open Source Community (led development efforts, coordinated contributors, and maintained core components)
 - Active contributor to collaborative technical projects and student initiatives
-

Additional Information

- Strong motivation to pursue graduate research in EEG-based human-computer interaction in Japan
- Committed to improving Japanese language skills for academic and daily life integration
- Strong interest in contributing to Aichi Prefecture's industrial ecosystem, particularly in robotics, automation, and intelligent systems
- Willing to work in Aichi-based industries after graduation and actively contribute to regional technological development